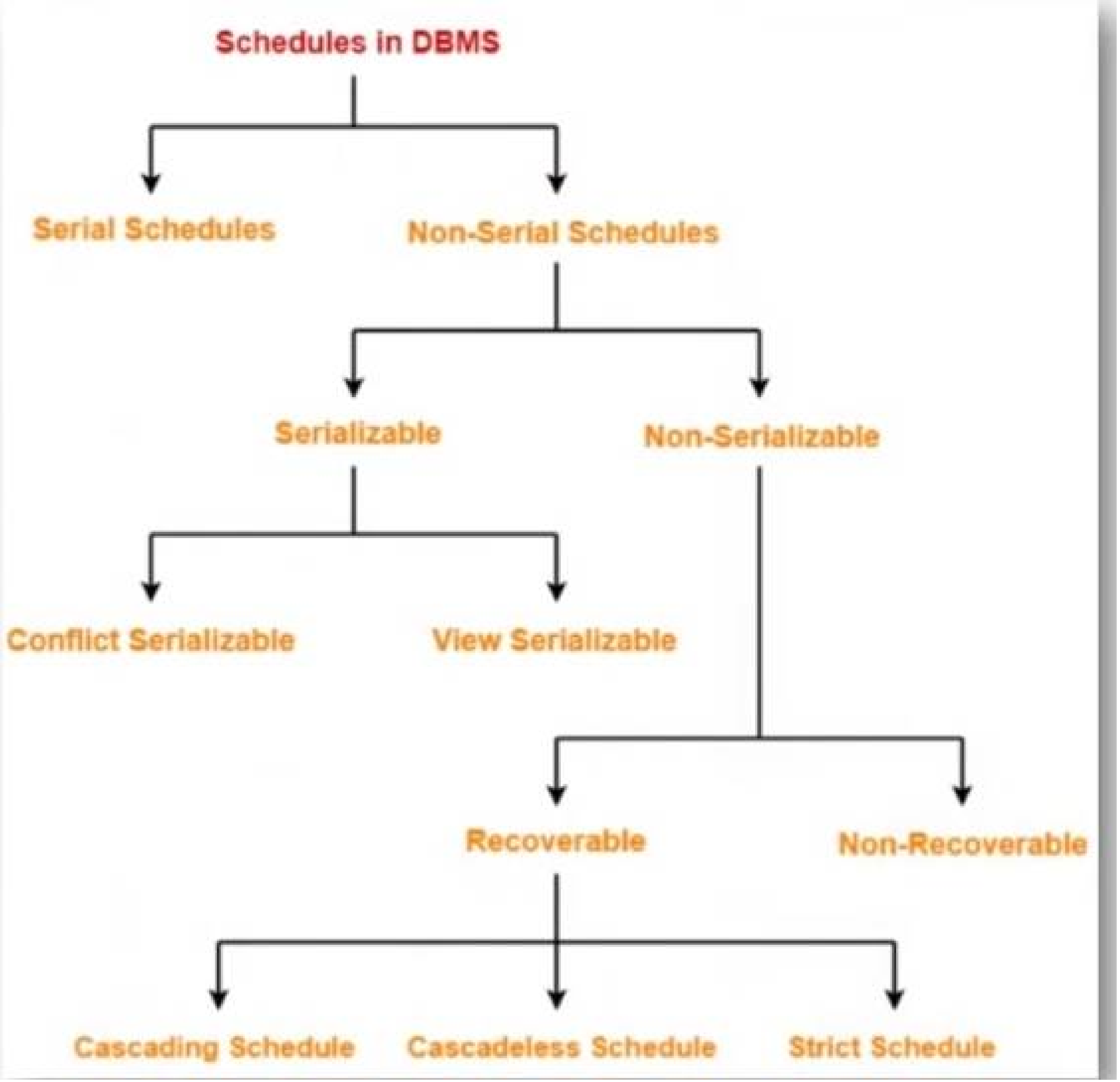

Schedules in DBMS
Serial Schedules
Non-Serial Schedules
Serializable
Non-Serializable
Conflict Serializable
View Serializable
Recoverable
Non-Recoverable
Cascading Schedule
Cascadeless Schedule
Strict Schedule

Serial & Non Serial Schedule

- Serial schedule is always a serializable schedule because transection executed one after another.
- In Non - serial schedule transections have performed parallel.
- Only Non-serial schedule needs to be checked for Serializability

Transaction T1	Transaction T2
R (A)	
W (A)	
R (B)	
W (B)	
Commit	
	R (A)
	W (B)
	Commit

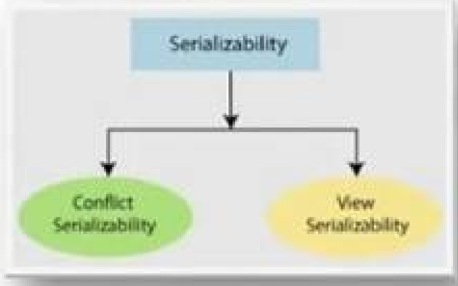
Serial Schedule: S

Transaction T1	Transaction T2
R (A)	
W (B)	
	R (A)
R (B)	
W (B)	
Commit	
	R (B)
	Commit

Non- Serial Schedule: S'

About Serializability

- Only Non-serial schedule needs to be checked for Serializability.
- Non-serial schedule is called a serializable **schedule if it can be converted to its equivalent serial schedule** OR **Result of serial & non -serial schedule is same.**
- Serializability identify correct non-serial schedules that will maintain the consistency of DB.
- Converted by using Conflict & View Serializable schedule.



Transaction T1	Transaction T2
R (A)	
W (A)	
R (B)	
W (B)	
Commit	
	R (A)
	W (B)
	Commit

Serial Schedule: S

Converted

Called Serializability

Transaction T1	Transaction T2
R (A)	
W (B)	
	R (A)
R (B)	
W (B)	
Commit	
	R (B)
	Commit

Non- Serial Schedule: S'

Conflict Serializability

- A schedule is called conflict serializability if after swapping of non-conflicting operations, it can transform into a serial schedule.

Non-serial schedule

T1	T2
Read(A) Write(A)	
	Read(A) Write(A)
Read(B) Write(B)	
	Read(B) Write(B)

Conflicting Pairs:

1. READ(a) - WRITE(a)
2. WRITE(a) - WRITE(a)
3. WRITE(a) - READ(a)

Conflicting Operations:

The two operations become conflicting if all conditions satisfy:

✓Both belong to separate transactions.

✓They have the same data item.

✓They contain at least one write operation.

Checking Whether a Schedule is Conflict Serializable Or Not

Step 1: Find all conflicting pairs.

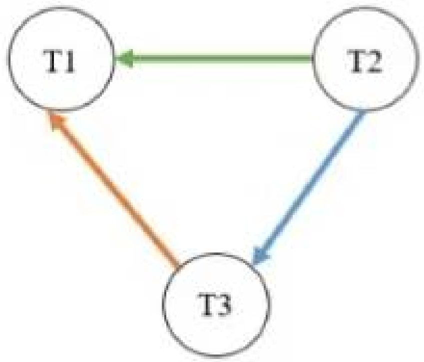
Step 2: Draw Precedence Graph.

T1	T2	T3
R(a)		
		R(b)
		R(a)
	R(b)	
	R(c)	
		W(b)
	W(c)	
R(c)		
W(a)		
W(c)		

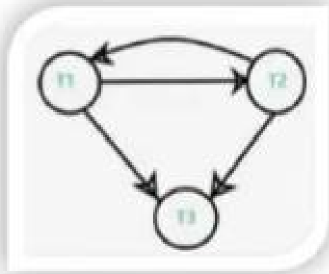
Given Schedule S

1. T3.R(a) – T1.W(a)
2. T2.R(b) – T3.W(b)
3. T2.R(c) – T1.W(c)
4. T2.W(c) – T1.R(c),W(C)

Step 1: Conflict Pairs



Step 2: Precedence Graph



Not Conflict Serializable Schedule

No Loops or Cycle Present in Graph
Therefore,
Schedule is Conflict Serializable Schedule

Conflicting Pairs:

1. READ(a) - WRITE(a)
2. WRITE(a) - WRITE(a)
3. WRITE(a) - READ(a)

Rules

Conflict Equivalent Serializability Example

Given:

T1	T2
R(a)	
W(a)	
R(b)	
	R(a)
	W(a)

Schedule S

T1	T2
R(a)	
W(a)	
	R(a)
	W(a)
R(b)	

Schedule S'

Find Out:
If S=S'Then
Conflict Equivalent

Step 1: Check Adjust Non Conflict Pairs as per Rules.
Step 2: If Present Swap it.

T1	T2
R(a)	
W(a)	
	R(a)
	W(a)
R(b)	

Schedule S'

Swap

T1	T2
R(a)	
W(a)	
	R(a)
R(b)	
	W(a)

Schedule S'

Swap

T1	T2
R(a)	
W(a)	
R(b)	
	R(a)
	W(a)

Schedule S'

Hence, S = S'
It is Conflict Equivalent

Rules:

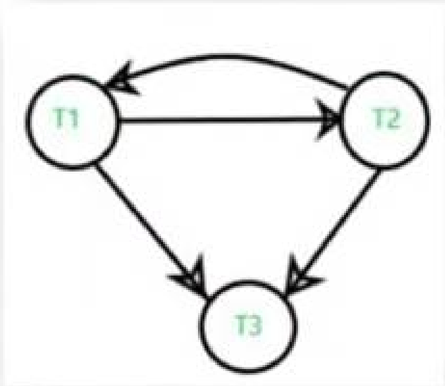
R(A) R(A)
R(A) W(A)
W(A) R(A)
W(A) W(A)
Conflict Pairs

R(B) R(A)
W(B) R(A)
R(B) W(A)
W(A) W(B)
Non Conflict Pairs

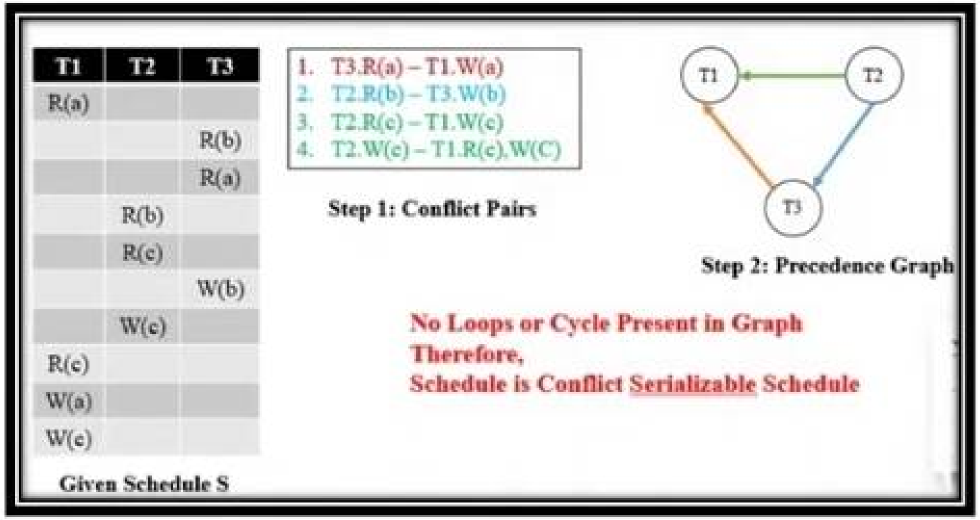
Why View Serializability?

- There may be some schedules that are Not Conflict-Serializable but gives a consistent result.
- Concept of Conflict-Serializability becomes limited when the Precedence Graph of a schedule contains a loop/cycle.
- In such a case we cannot predict whether a schedule would be consistent or inconsistent

T1	T2	T3
R(a)		
	W(a)	
W(a)		
		W(a)



Loops or Cycle Present in Graph
Therefore,
Schedule is Non Conflict Serializable Schedule



T1	T2	T3
R(a)		
		R(b)
		R(a)
	R(b)	
	R(c)	
		W(b)
	W(c)	
R(c)		
W(a)		
W(c)		

Given Schedule S

1. T3.R(a) – T1.W(a)
2. T2.R(b) – T3.W(b)
3. T2.R(c) – T1.W(c)
4. T2.W(c) – T1.R(c),W(C)

Step 1: Conflict Pairs

Step 2: Precedence Graph

No Loops or Cycle Present in Graph
Therefore,
Schedule is Conflict Serializable Schedule

Discussed in Part 2:
Conflict Serializable Schedule

View Serializability

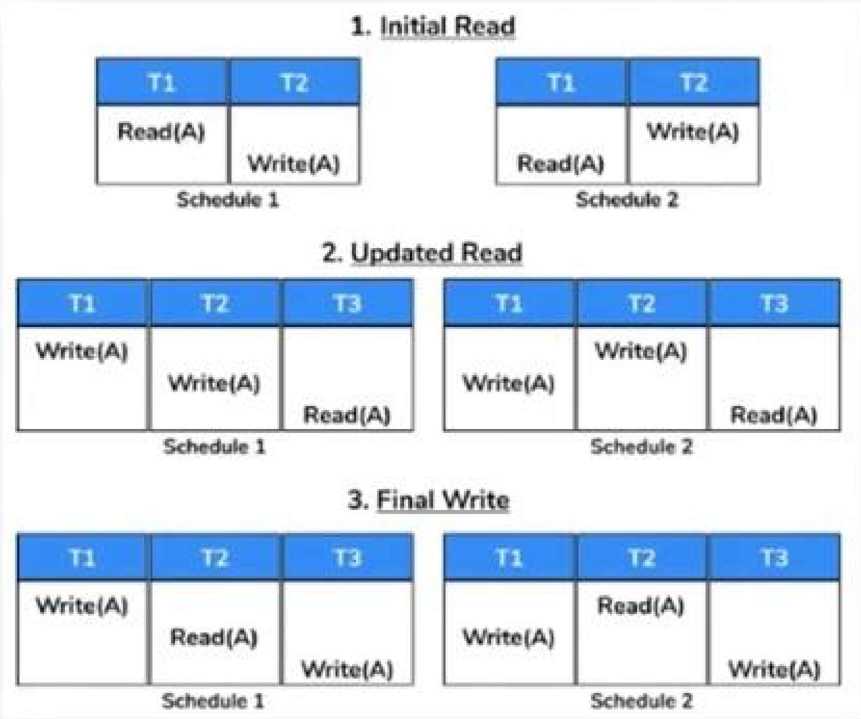
- A schedule will view serializable if it is view equivalent to a serial schedule.
- **All conflict serializable schedules are said to be view serializable.**
- **All view serializable schedules might or might not be conflict serializable.**

Conditions:

✓Initial readers should be similar for entire data items.

✓The write-read sequence should be identical.

✓Final writers should be identical for whole data items.

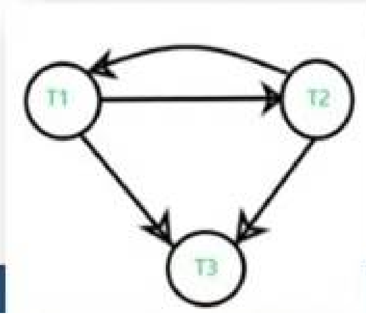


View Serializability Example

Given a= 100

T1	T2	T3
R(a) = 100		
	W(a) a = a - 40 60	
W(a) a = a - 40 20		
		W(a) a = a - 20 0

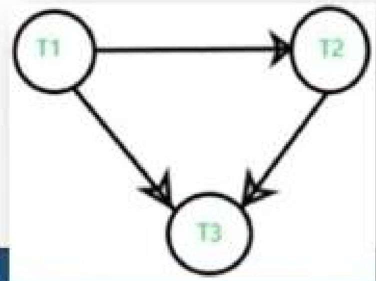
S: Non Conflict Serializable Schedule



Swap some Transections
As per conditions

T1	T2	T3
R(a) = 100		
W(a) a = a - 40 60		
	W(a) a = a - 40 20	
		W(a) a = a - 20 0

S': Conflict / View Serializable Schedule



S = S'
Final Result is Same
Therefore,
View Equivalent Schedule